BLEND DOOR ACTUATOR (Continued)

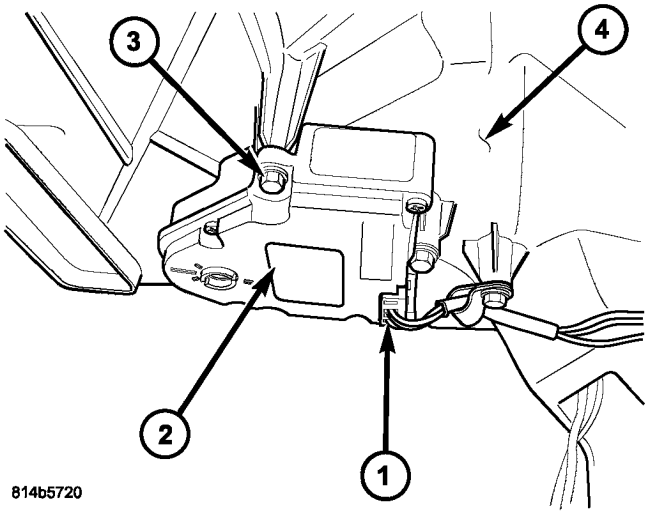

Fig. 23 Blend Door Actuator

1 - WIRE HARNESS CONNECTOR
2 - BLEND DOOR ACTUATOR
3 - SCREWS (3)
4 - HVAC HOUSING

INSTALLATION

(1) Position the blend door actuator onto the bottom of the HVAC housing. If necessary, rotate the actuator slightly to align the actuator output shaft with the blend door linkage.

(2) Install the three screws that secure the blend door actuator to the HVAC housing. Tighten the screws to 2 N·m (17 in. lbs.).

(3) Connect the HVAC wire harness connector to the blend door actuator.

(4) Reconnect the battery negative cable.

BLOWER MOTOR RELAY

DESCRIPTION

The blower motor relay (Fig. 24) is a International Standards Organization (ISO)-type relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal functions and patterns. The blower motor relay is a electromechanical device that switches battery current from a fuse in the power distribution center (PDC) directly to the blower motor. The blower motor relay is energized when the relay coil is provided a voltage signal by the ignition switch.

The blower motor relay is mounted to the HVAC housing, directly behind the glove box.

OPERATION

The ISO-standard blower motor relay is an electromechanical switch that uses a low current input from the ignition switch to control the high current output to the blower motor. The movable, common feed relay contact is held against the fixed, normally closed relay contact by spring pressure. When the electromagnetic relay coil is energized, it draws the movable common feed relay contact away from the fixed, normally closed relay contact and, holds it against the fixed, normally open relay contact. This action allows high current to flow to the blower motor.

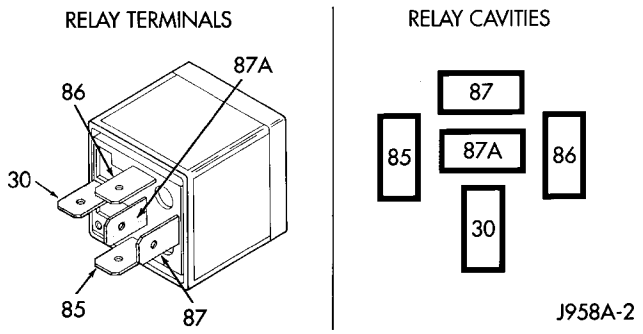

Fig. 24 Blower Motor Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

When the relay coil is de-energized, spring pressure returns the movable relay contact back against the fixed, normally closed contact point. The resistor or diode is connected in parallel with the relay coil, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

The blower motor relay terminals are connected to the vehicle electrical system through a connector receptacle mounted to the HVAC housing. The inputs and outputs of the blower motor relay include:

- The common feed terminal (4) receives battery current from a fuse in the PDC at all times.
- The coil ground terminal (2) is connected to ground at all times.
- The coil battery terminal (5) receives battery current from a fuse in the fuse block through a fused ignition switch output (run) circuit only when the ignition switch is in the On position.
- The normally open terminal (1) provides battery current to the blower motor through the blower motor relay only when the blower motor relay coil is energized.
- The normally closed terminal (87A) is not connected to any circuit in this application, but provides